

Loops: Repeating Instructions in Programming

Introduction

- This video builds on the "Introduction to Programming" lesson.
- The main topic is how to give **instructions to a computer** to perform tasks repeatedly.
- Computers are excellent at repeating instructions, similar to how a person counts repetitions.

The Need for Loops

- **Problem 1** : Print "Hello World" **5 times** .
- **Inefficient Solution** : Writing `print("Hello World")` **5 separate times** .
- This approach is repetitive and not scalable for many repetitions.
- **Loops** provide an efficient way to repeat instructions without writing the same code multiple times.

The 'for' Loop

- A **'for' loop** is a common type of loop used to iterate over a sequence (like numbers in a range) or repeat a task a specific number of times.

Basic 'for' Loop Structure

- Syntax: `for i in range(5):`
- `i` is a **variable** that changes with each repetition, often called an **iterator** .
- `range(5)` specifies that the loop will run **5 times** .
- `range(5)` generates numbers **0, 1, 2, 3, 4** by default.
- **Indentation** is crucial in Python; code indented under the `for` line is part of the loop.
- Example: `print(i)` inside the loop would print `0, 1, 2, 3, 4` .

'range()' Function Variations

- `range(stop)` : Generates numbers from **0** up to (but not including) `stop` .
- Example: `range(5)` generates `0, 1, 2, 3, 4` .
- `range(start, stop)` : Generates numbers from `start` up to (but not including) `stop` .
- **Problem 2** : Print numbers from **1 to 10** .
- **Solution** : `for i in range(1, 11):` will print `1, 2, ..., 10` .
- `range(start, stop, step)` : Generates numbers from `start` up to (but not including) `stop` , incrementing by `step` .
- **Problem 3** : Print **even numbers** from **2 to 10** .
- **Solution** : `for i in range(2, 11, 2):` will print `2, 4, 6, 8, 10` .
- The `step` value defines the increment for each iteration.

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Note

- **These notes were cut off** because the video is extremely long. Consider asking for the video in shorter parts.